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(54) **METHODS AND SYSTEMS FOR DETECTING
UNSEALED ACCESS POINTS IN AN INFANT
CARE STATION**

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E04H 15/32 (2006.01)

(57)

ABSTRACT

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(2013.01); **E04H 15/32** (2013.01)

An infant care station can include sensors to detect an environmental characteristic of the infant care station, an access point to access a microenvironment of the infant care station, and a processor to obtain a sealed measurement for the infant care station with the access point in a sealed position. The processor can also obtain the environmental characteristic from sensors monitoring the microenvironment. Additionally, the processor can determine a difference between the environmental characteristic and the sealed measurement and generate an alert indicating an access point sealing issue based on the difference exceeding a predetermined threshold.

(58) **Field of Classification Search**

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See application file for complete search history.

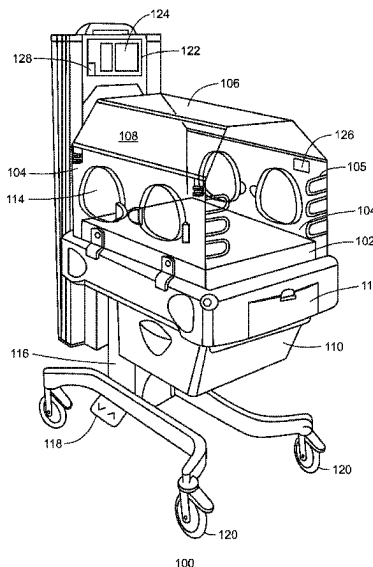
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10 Claims, 5 Drawing Sheets



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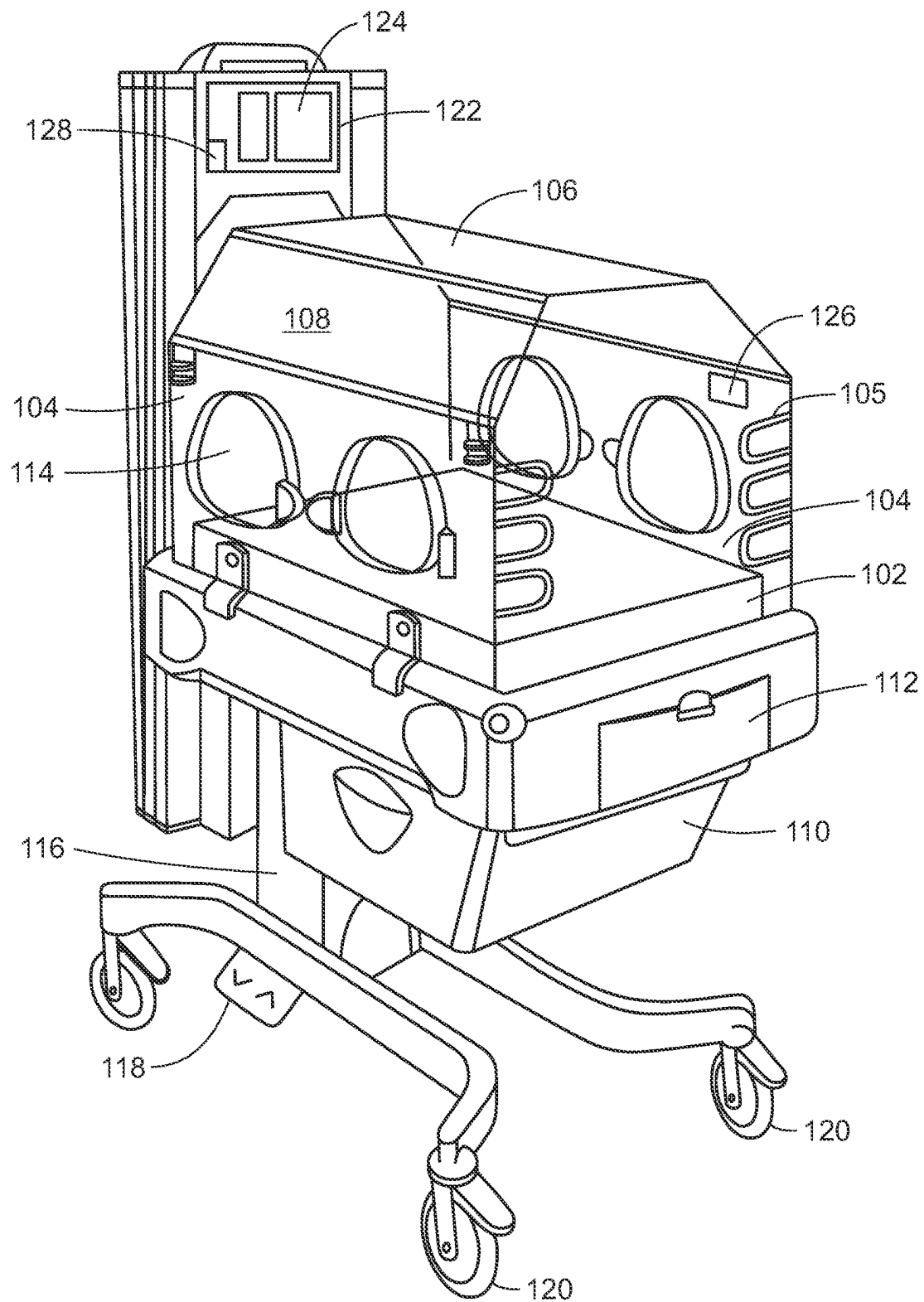
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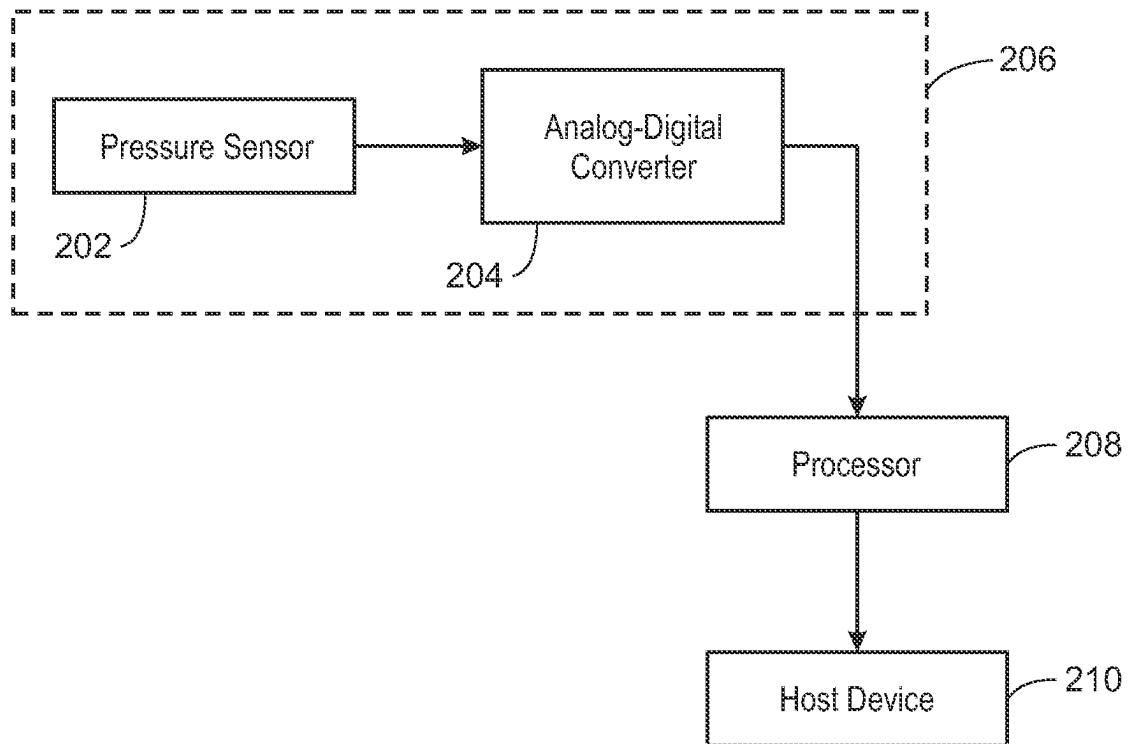
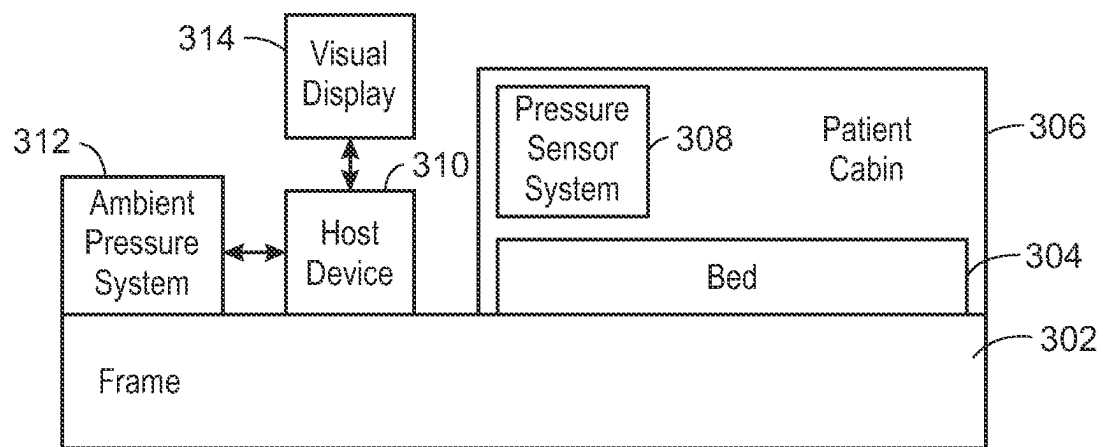
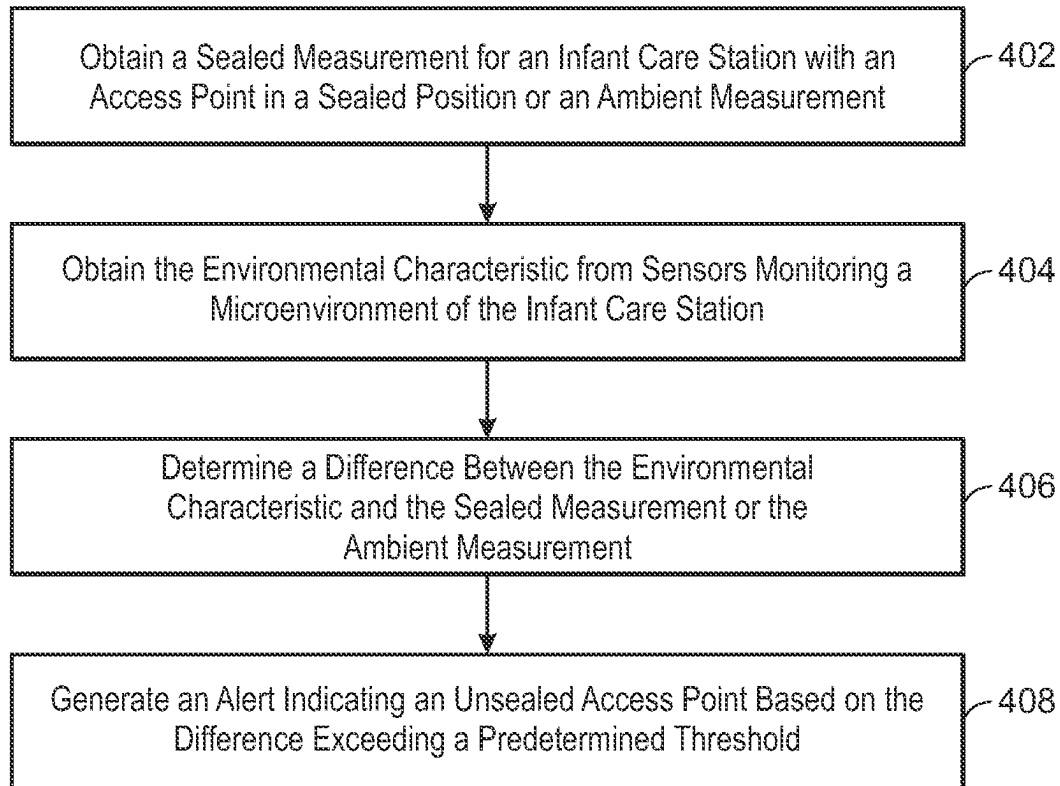


FIG. 2



300
FIG. 3



400
FIG. 4

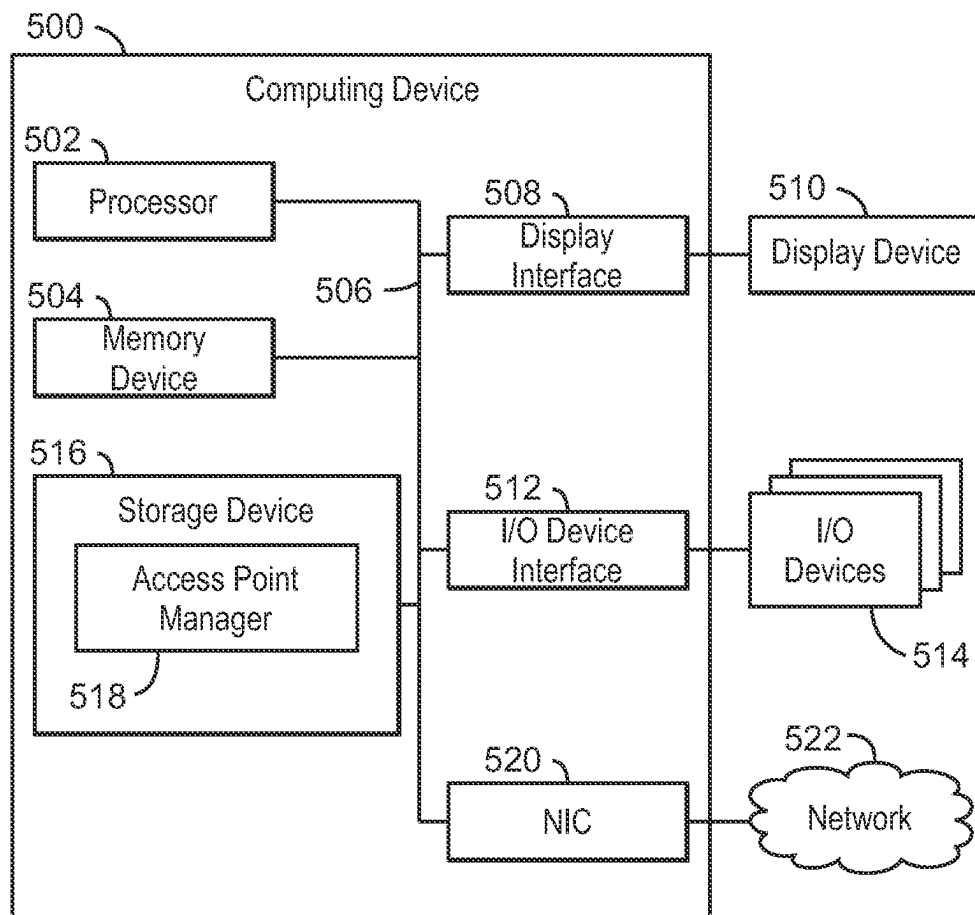


FIG. 5

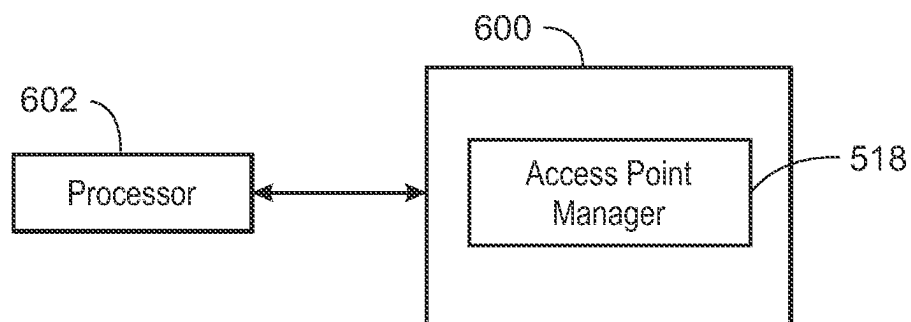


FIG. 6

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METHODS AND SYSTEMS FOR DETECTING UNSEALED ACCESS POINTS IN AN INFANT CARE STATION

BACKGROUND

The present disclosure generally relates to infant care stations, and more specifically to detecting sealed or unsealed access points in an enclosure of an infant care station.

Some neonates are not physiologically well enough developed to be able to survive without special medical attention. A frequently used medical aid for such infants is the incubator. The primary objective of the incubator is to provide an environment which will maintain the neonate at a minimum metabolic state thereby permitting as rapid physiological development as possible. Neonatal incubators create a microenvironment that is thermally neutral where a neonate can develop. These incubators typically include a humidifier and a heater and associated control system that controls the humidity and temperature in the neonatal microenvironment. The humidifier comprises a device that evaporates an evaporant, such as distilled water, to increase relative humidity of air within the neonatal microenvironment. The humidifier is typically controllable such that the amount of water, or water vapor, added to the microenvironment is adjustable in order to control the humidity to a desired value. The heater may be, for example, an air heater controllable to maintain the microenvironment area to a certain temperature. Radiant warmers may be used instead of incubators for some neonates where less environmental control is required. In still other embodiments, hybrid incubator/radiant warming systems may be utilized.

Since the microenvironment is accurately controlled in a neonatal care system, the care system includes an enclosure that is sealed as best possible to help maintain the controlled microenvironment. Such an enclosure will typically include four sidewalls or side panels and a top hood that surround an infant support platform. Typically, one or more of the side panels can include access points, such as porthole doors, and a removable top, among others, that enable clinicians to access neonates in the microenvironment. In some examples, the access points may not be fully closed or sealed, which can alter the conditions of the microenvironment.

SUMMARY

This Summary is provided to introduce a selection of concepts that are further described below in the Detailed Description. This Summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter.

An infant care station can include sensors to detect an environmental characteristic of the infant care station, an access point to access a microenvironment of the infant care station, and a processor to obtain a sealed measurement for the infant care station with the access point in a sealed position. The processor can also obtain the environmental characteristic from sensors monitoring the microenvironment of the infant care station. Additionally, the processor can determine a difference between the environmental characteristic and the sealed measurement and generate an alert indicating an access point sealing issue based on the difference exceeding a predetermined threshold. The access point

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sealing issue, as referred to herein, can indicate an unexpected open or unsealed access point, or an unexpected, sealed access point.

In some examples, the access point can include a canopy, a porthole door, equipment access point, or a combination thereof. In one aspect, the processor can transmit the alert to an external computing device or a remote display device. In some examples, the processor can display the alert in a user interface coupled to the system.

In one aspect, the environmental characteristic can include a pressure value, a humidity value, an oxygen value, a temperature control loop value, or a combination thereof. In some examples, the sensors can include a pressure sensor, an oxygen sensor, a humidity sensor, or a combination thereof.

In one aspect, the sensors are coupled to an enclosure of the system, wherein the enclosure supports the microenvironment. In some examples, the alert can include an audio feedback, haptic feedback, or visual feedback representing a message that the at least one access point is unsealed. In some examples, the infant care station can include an ambient sensor, wherein the ambient sensor can detect an ambient measurement, and wherein the processor can generate a second alert in response to detecting a difference between the ambient measurement and the environmental characteristic that exceeds an ambient threshold. In some examples, the sensors can include a pressure sensor within the microenvironment and the environmental characteristic can include a pressure value obtained from the microenvironment by the pressure sensor.

In some examples, a method can include obtaining an ambient measurement from one or more ambient sensors for an environment proximate to an infant care station. The method can also include obtaining an environmental characteristic from one or more microenvironment sensors and determining a difference between the environmental characteristic and the ambient measurement exceeds a predetermined threshold. Additionally, the method can include generating an alert indicating an access point sealing issue in the infant care station based on the difference exceeding the predetermined threshold. In one aspect, the method can include preventing the alert from being generated in response to detecting a user in proximity sensor data within a predetermined distance from the infant care station.

In one aspect, a non-transitory machine-executable medium for detecting an access point sealing issue in an infant care station can include a plurality of instructions that cause a processor to obtain a sealed measurement for the infant care station with the at least one access point in a sealed position, wherein the at least one access point comprises a canopy, a porthole door, an equipment access point, or a combination thereof. The plurality of instructions can also cause the processor to obtain an environmental characteristic from the one or more sensors and determine a difference between the environmental characteristic and the sealed measurement exceeds a predetermined threshold. In some examples, the plurality of instructions can also cause the processor to generate an alert indicating an access point sealing issue in the infant care station based on the difference exceeding the predetermined threshold.

Various other features, objects, and advantages of the invention will be made apparent from the following description taken together with the drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

The drawings illustrate the best mode presently contemplated of carrying out the disclosure. In the drawings:

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FIG. 1 is a perspective view of an example infant care station in accordance with one example;

FIG. 2 is a block diagram of an example of a pressure sensor system within an infant care station;

FIG. 3 depicts a block diagram of an example system for detecting an access point sealing issue in an infant care station;

FIG. 4 depicts a process flow diagram of an example method for detecting an access point sealing issue in an infant care station;

FIG. 5 is a block diagram of an example of a computing device that can detect an access point sealing issue of an infant care station; and

FIG. 6 depicts a non-transitory machine-executable medium with instructions that can detect an access point sealing issue in an infant care station.

The drawings illustrate specific aspects of the described components, systems and methods for providing a neonatal incubator system. Together with the following description, the drawings demonstrate and explain the principles of the structures, methods, and principles described herein. In the drawings, the thickness and size of components may be exaggerated or otherwise modified for clarity. Well-known structures, materials, or operations are not shown or described in detail to avoid obscuring aspects of the described components, systems and methods.

DETAILED DESCRIPTION

Embodiments of the present disclosure will now be described, by way of example, with reference to FIGS. 1-6. Infant care stations can provide microenvironments for infant patients receiving medical care. Infant care stations, as referred to herein, can include incubators, warmers, or devices that support one or more features of incubators and warmers. In some examples, the infant care stations can enable clinicians to access the patient by opening one or more access points. An access point, as referred to herein, includes porthole doors that reside within one or more walls of the infant care stations, removable canopies of infant care stations, equipment access points, and the like. For example, a clinician may disengage any suitable latch coupled to the porthole doors to open the porthole doors and access a patient residing within an infant care station. However, porthole doors can be accidentally left open, which can result in unexpected conditions within the microenvironment of the infant care station. Also, radiant heaters of an infant care station can be improperly engaged when an infant care station transitions from a warmer with an open top or canopy to an incubator with a closed canopy. When the radiant heater remains engaged for an extended period of time with the canopy of the infant care station in a closed position, unexpected conditions can arise in the microenvironment.

Techniques described herein enable an infant care station to detect when an access point is sealed or unsealed. In some examples, an infant care station can include one or more sensors (also referred to herein as microenvironment sensors) that can measure, obtain, or otherwise detect an environmental characteristic of the microenvironment. The environmental characteristic, as referred to herein, can include a pressure level, an oxygen level, a humidity level, or a combination thereof obtained from the microenvironment of the infant care station. In some examples, the infant care station can monitor the environmental characteristic and determine when an access point is left unsealed or open based on a previously obtained sealed measurement from the

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microenvironment of the infant care station. The sealed measurement, as referred to herein, can indicate an expected pressure, oxygen level, humidity level, and the like, within the microenvironment when the porthole doors are in a sealed or closed position. Alternatively, the infant care station can detect an ambient characteristic or measurement, such as a pressure, oxygen level, or humidity level, from the surrounding environment outside of the infant care station. In some examples, the infant care station can compare the ambient characteristic to the environmental characteristic to determine if an access point is sealed or unsealed. In some examples, a sealed or unsealed access point refers to a gap between an access point and an infant care station exceeding a predetermined threshold. For example, a sealed access point can refer to a canopy or a porthole door, among other access points, that is positioned proximate walls of an infant care station such that a gap exists between the walls and the canopy or porthole door. In some examples, a gap that is smaller than a predetermined threshold represents a sealed access point and a gap greater than a predetermined threshold represents an unsealed access point.

An advantage that may be realized by the sealed access point detection feature in the practice of some examples of the described systems and techniques is an additional safety mechanism to prevent an access point from remaining disengaged in an unsealed or open position for an extended period of time. The techniques herein can also prevent unexpected conditions within the microenvironment by detecting when an infant care station has transitioned from a warmer to an incubator and determining if a heater has turned off following the transition. Accordingly, techniques herein can prevent an unexpected alteration of the temperature, humidity, oxygen level, and the like, within the microenvironment of an infant care station. Techniques for detecting sealed or unsealed access points are described in greater detail below in relation to FIGS. 1-6.

FIG. 1 is a perspective view of an example infant care station in accordance with one example. In the example of FIG. 1, an infant care station is depicted in which the infant care station is an incubator 100. The incubator 100 includes a horizontal surface 102 that is configured to support an infant patient (not depicted). It is to be understood that the incubator 100 may have the ability or control to move, rotate, or incline the horizontal surface 102; however, it will be understood that the horizontal surface 102 will generally remain horizontal such as to minimize movement of the infant patient within the incubator 100 due to gravity.

One or more walls 104 extend generally vertically from the horizontal surface 102. In the embodiment depicted in FIG. 1 of the incubator 100, four walls extend vertically from the horizontal surface 102 to define the rectangular shape of the incubator 100. However, it will be understood that in alternative examples, various numbers of walls 104 may be used to define the incubator into various geometric shapes which may include, but are not limited to, circles or hexagons. The incubator 100 can further include a canopy 106 that extends over the horizontal surface 102. In some examples, the canopy 106 can include multiple components or surfaces, or the canopy may be curved or domed in shape.

While the incubator of FIG. 1 is depicted with the horizontal surface 102, walls 104, and canopy 106 being connected, it will be understood that in alternative examples, including those described in greater detail herein, the horizontal surface 102, walls 104, and canopy 106 may be individual components that also may be moveable with respect to each other. For example, the canopy 106 can transition from a closed position to an open position in

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which any suitable portion of the canopy **106** is raised away from the walls **104** to allow the microenvironment to be exposed to the surrounding environment of the incubator **100**.

The horizontal surface **102**, walls **104**, and canopy **106** can define a microenvironment **108** contained within these structures. In some examples, the incubator **100** is configured such that the microenvironment **108** surrounds the infant patient (not depicted) such that the infant patient is only exposed to a controlled combination of environmental characteristics or conditions (temperature, humidity, O₂ concentration, etc.) selected by a clinician to promote the health and wellbeing of the infant patient. In some examples, the walls **104** further include arm portholes **114** that permit a clinician access into the microenvironment **108**. The walls **104** can also include any number of equipment access points **105**. The equipment access points **105** can include grommet seals (or any other suitable seals) for cables, harnesses, and other medical equipment access holes. In some examples, seals and other equipment can fall out from the equipment access points **105** or the seals and other equipment can be damaged when equipment hoses or cables are removed.

In some examples, the incubator **100** includes a base **110** that houses a convective heater **112**. The convective heater **112** is operated such that air is drawn into the incubator **100**, at which point the air may be filtered or sterilized in another manner, including the use of UV light before being passed by heating coils (not depicted) to heat the air to a target or set point temperature. The sterilized and heated air is blown into the microenvironment **108** through vents (not depicted) which are arranged along the walls **104**. As is also known, the air may be entrained with supplemental gasses such as oxygen or may have added humidity such as to control these conditions within the microenvironment **108**.

Examples of the incubator **100** further include a pedestal **116** connected to the base **110**. The pedestal **116** includes mechanical components (not depicted), which may include, but are not limited to, servo motors, rack and pinion systems, or screw gear mechanisms that are operable by foot pedals **118** to raise or lower the base **110**, effectively raising or lowering the position of the infant patient (not depicted) in relation to the clinician. The incubator **100** may be moveable by wheels or casters **120** connected to the pedestal **116**.

The example of the incubator **100** depicted in FIG. **1** includes a graphical display **122** that is mounted to a wall, the base **110**, or the canopy **106** of the incubator **100** at a position external to the microenvironment **108**. The graphical display **122** is operated by a processor to present a graphical user interface (GUI) **124**. In the example illustrated, the graphical display **122** is a touch-sensitive graphical display and the GUI **124** is configured to specifically respond to inputs made by a clinician received through the touch-sensitive graphical display. During normal operation, the touch-sensitive graphical display **122** and touch-sensitive configured GUI **124** are used to control various functions of the incubator **100**. The GUI **124** presents a variety of information, such as the air temperature and alarm indications. In some examples, the alarm indications can provide a message indicating an access point is unsealed or open, a change in environment characteristics, or a warning that a heater is still operational after the canopy **106** has been closed, among others.

In some examples, the walls **104** of the incubator **100** can be opened or closed to enable a clinician to access a patient residing in the incubator **100**. For example, the walls **104** can serve as doors that open and close to either remove a patient from the incubator **100** or to place a patient into the

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incubator **100**. As described in greater detail below in relation to FIGS. **2-6**, the walls **104** can include any number of access points, such as portholes **114** covered by porthole doors, that enable access to a patient residing in a microenvironment of the incubator **100**.

In some examples, the incubator **100** can include any number of sensors as part of a sensor system **126**. The sensor system **126** can include pressure sensors, oxygen sensors, humidity sensors, and the like. In some examples, the sensor system **126** is connected to a host device **128** that controls the GUI **124**. The sensor system **126** can transmit sensor data to the host device **128** and the host device **128** can determine if any access points, such as the canopy **106** or portholes **114**, of the incubator **100** are unsealed or open. In some examples, the sensor system **126** can transmit sensor data indicating environmental characteristics of a microenvironment to a host device **128** using any suitable wired or wireless transmission protocol. The host device **128** can determine if the access points are unsealed or open based on generating a difference using the sensor data in real-time and sealed measurements previously obtained from the incubator **100** or an ambient sensor measurement as discussed in greater detail below in relation to FIG. **3**.

FIG. **2** is a block diagram of an example of a pressure sensor system within an infant care station. In some examples, the pressure sensor **202** of an infant care station is used to determine an environmental characteristic of a microenvironment of an infant care station. The pressure sensor **202** can be coupled to an analog-to-digital converter **204**, or the pressure sensor **202** and the analog-to-digital sensor **204** can be combined to form an integrated pressure sensor **206**. The analog-to-digital converter **204** or the integrated pressure sensor **206** can transmit sensor data to a processor **208** at any suitable time interval such as one second, 10 seconds, one minute, or the like. The processor **208** can transmit the sensor data to a host device **210**, or the host device **128** of FIG. **1**, among others. In some examples, either the processor **208** or the host device **210** can determine if an access point is open or unsealed by determining if a measurement from the microenvironment of an infant care station exceeds a predetermined difference with an ambient sensor value or a sealed measurement previously obtained within the infant care station. In some examples, the processor **208** can transmit sensor data or data calculated based on the sensor data to the host device **210** using any suitable wired or wireless protocol.

In some examples, an infant care station, such as the incubator **100** of FIG. **1**, can include any number of pressure sensors **202**, among other sensors. The host device **210** can obtain sensor data from the pressure sensors **202** and any other sensors and determine if an infant care station has one or more access points that are either in an open or closed position. Additionally, in some examples, the analog-to-digital converter **204** or the integrated pressure sensor **206** can include logic to pre-process the sensor data to detect incorrect sensor data and the like.

FIG. **3** depicts a block diagram of an example system for detecting an access point sealing issue in an infant care station. In some examples, the infant care station **300** can include a frame **302** that supports a bed **304** within a patient cabin **306**. The patient cabin **306** can include four or more walls and a canopy, among other components that enable a microenvironment to form within the patient cabin **306**. For example, the patient cabin **306** can maintain a constant humidity, oxygen level, temperature, and the like for a patient residing on the bed **304**. In some examples, a pressure sensor system **308** is coupled to the patient cabin

such that the pressure sensor system **308** can determine a pressure of the microenvironment.

In some examples, the pressure sensor system **308** can periodically or continuously transmit pressure sensor data to a host device **310**. The host device **310** can also obtain ambient sensor data or ambient measurements from ambient pressure sensors or an ambient pressure system **312** proximate to the host device **310**. The ambient pressure system **312** can provide the pressure values from the environment outside of the microenvironment. In some examples, the pressure values obtained, detected, or otherwise received from the ambient pressure system **312** can indicate an altitude of an infant care station **300**. For example, a lower pressure level can indicate that the infant care station **300** resides at a higher altitude.

In some examples, the host device **310** can obtain an environmental characteristic, such as a pressure of the microenvironment, from the pressure sensor system **308** and determine whether a difference between the environmental characteristic and the ambient measurement exceeds a predetermined threshold. If the difference does not exceed the predetermined threshold, the host device **310** may continue to monitor the microenvironment without generating an alert. If the difference does exceed the predetermined threshold, the host device **310** can generate an alert indicating an unsealed access point in the infant care station based on the difference exceeding the predetermined threshold.

In some examples, the alert can be provided to a user with the visual display **314**, using haptic feedback, or an audio feedback, among others. In some examples, the visual display **314** can be coupled to the host device **310** or the visual display **314** can reside in a remote location. The alert can include a message that the at least one access point is unsealed or that a canopy of an infant care station **300** has transitioned to a closed position from an open position.

In some examples, the host device **310** can obtain environmental characteristics from any number of sensors coupled to the infant care station **300**. The sensors can monitor the oxygen level, humidity level, and the like, from within the microenvironment, or the sensors can monitor components that control the temperature of the microenvironment. For example, the host device **310** can detect a decrease in oxygen level of a microenvironment of an infant care station **300**, which indicates an open access point, such as a porthole door or a canopy. The host device **310** can also detect an increase or decrease in power consumption by a convective heater, convective heater fan, among other heater components, of an infant care station **300**.

It is to be understood that the block diagram of FIG. 3 is not intended to indicate that the infant care station **300** is to include all of the components shown in FIG. 3. Rather, the infant care station **300** can include fewer or additional components not illustrated in FIG. 3 (e.g., additional memory components, embedded controllers, additional modules, additional network interfaces, additional sensor devices, etc.).

FIG. 4 depicts a process flow diagram of an example method for detecting an access point sealing issue in an infant care station. The method **400** can be implemented with any suitable infant care station, such as the incubator system **100** of FIG. 1 or the infant care station **300** of FIG. 3, among others.

At block **402**, the method **400** can include obtaining, detecting, or otherwise receiving a sealed measurement for an infant care station with at least one access point in a sealed position or obtaining, detecting, or otherwise receiving an ambient measurement. In some examples, the sealed

measurement can be obtained as a predetermined value detected using one or more sensors of an infant care station when the access points are sealed or closed. The sensors can detect any suitable sealed measurement such as a pressure level within the microenvironment of the infant care station, a humidity level within the microenvironment of the infant care station, an oxygen level within the microenvironment of the infant care station, a temperature control loop value, or any combination thereof. The temperature control loop value can indicate any suitable measurement used to control a fan motor, a heater duty cycle, a radiant heater fan speed, or the like. The temperature control loop value can be obtained with any suitable control loop sensor that monitors the power consumption of heating components within an infant care station. In some examples, the temperature control loop value can indicate a fan speed, heater elements on or off times, an increased wattage for a radiant heater, and an increased wattage for a humidifier heater, among others. The temperature control loop value can represent when a heater of an infant care station is providing heat to a microenvironment. In some examples, the temperature control loop value is modified in response to detecting a temperature variation in the microenvironment. For example, an increase or a decrease of a temperature of the microenvironment can cause a heater of an infant care station to provide a modified amount of heat so that the temperature of the microenvironment is maintained within a predetermined temperature range.

In some examples, the method **400** can detect, determine, or otherwise obtain an ambient measurement instead of, or in addition to, obtaining a sealed measurement. The ambient measurement can be measured or obtained by an ambient sensor monitoring the environment proximate an infant care station as described in greater detail above in relation to FIG. 3. The ambient measurement can indicate a pressure level or humidity level, among others, of the environment surrounding an infant care station.

At block **404**, the method **400** can include obtaining the environmental characteristic from one or more microenvironment sensors monitoring the microenvironment. The microenvironment sensors can include pressure sensors, humidity sensors, oxygen sensors, and the like. In some examples, the microenvironment sensors can be placed at any suitable location within the enclosure or cabin of an infant care station to monitor environmental characteristics such as pressure levels, oxygen levels, and humidity levels, among others. For example, pressure sensors, oxygen sensor, humidity sensors, and the like, can be placed proximate to porthole doors or canopies, among other access points. In some examples, one or more pressure sensors can be included within an infant care station to detect a set of pressure values from different locations within the microenvironment. The set of pressure values can enable a host device to determine an average pressure value or a difference in pressure values. The difference in pressure values can be used to calculate or compute a rate of change in the pressure values detected by each of the pressure sensors, which can indicate the pressure sensor that is closest to an unsealed or open access point. In some examples, multiple pressure sensors, oxygen sensors, humidity sensors, or other microenvironment sensors, can be used to determine an open or unsealed access point by detecting the microenvironment sensor that is monitoring the faster rate of change as compared to other microenvironment sensors in the infant care station.

The method **400** can also include obtaining sensor data values from different types of sensors monitoring the

microenvironment of an infant care station. In some examples, the sensors can obtain sensor data directly from the microenvironment or indirectly by monitoring the operation of heaters and other components of the infant care station. For example, the sensors can indirectly determine a temperature change or temperature stability of the microenvironment of an infant care station by monitoring sensor data from control loop sensors that detect, determine, or otherwise obtain the temperature control loop values that represent the operation of radiant heaters, among other components, of the infant care station. In some examples, the sensors can detect sensor data indicating that a control loop for an infant care station is becoming more active, which can represent a perturbation of the microenvironment of the infant care station via the increased control loop activity.

At block 406, the method 400 can include determining a difference between the environmental characteristic and either the sealed measurement or the ambient measurement. As discussed above in relation to block 402, the sealed measurement can be obtained from the microenvironment of the infant care station with the access points in a sealed or closed position and the ambient measurement can be obtained from any suitable ambient sensor. The difference between either the environment characteristic and the sealed measurement or the environmental characteristic and the ambient measurement can represent a step change in pressure, oxygen level, humidity, and the like, within the microenvironment of an infant care station. The difference can indicate that one or more access points may have been unsealed or opened for a period of time. For example, the difference can indicate that a porthole door, a canopy, or any other suitable access point of an infant care station has been opened. In some examples, an open access point can enable warm air from the microenvironment to exit the infant care station into the surrounding environment. The difference can also indicate that the infant care station has transitioned from an open canopy position to a closed canopy position.

In some examples, the sensors can also include global positioning system (GPS) coordinates that map the location of the infant care station to a known altitude. The known altitude can be used when determining if the environmental characteristic exceeds a predetermined threshold. For example, the altitude can be used to determine an expected pressure of the environment surrounding the infant care station, an expected baseline pressure of the microenvironment in an unsealed or open position, or the like.

At block 408, the method 400 can include generating an alert indicating an access point sealing issue based on the difference exceeding a predetermined threshold. The access point sealing issue, as referred to herein, can indicate an unexpected open or unsealed access point or an unexpected, sealed access point. For example, the access point sealing issue can indicate an open porthole door or a closed canopy, among others. In some examples, the alert can indicate an amount of time any number of access points have been open, whether the amount of time an access point has been open exceeds a predetermined threshold, a pressure value, an oxygen value, a humidity value, a rate of change in a set of pressure values, oxygen values, or humidity values, or a delta value representing a change in pressure values, oxygen values, or humidity values within a predetermined period of time, among others. In some examples, the alert can indicate values from two or more different types of sensors have exceeded predetermined thresholds. For example, the alert can indicate a difference between a measured pressure value and a predetermined sealed pressure value for the infant care

station has exceeded a first threshold and a difference between a measured oxygen level and a predetermined sealed oxygen level for the infant care station has exceeded a second threshold. In some examples, any number of types of sensors can detect different sensor values and the alert can indicate if any of the different sensors obtain sensor values that exceed one or more thresholds. For example, the sensor devices can include sensors that monitor the pressure, humidity, and oxygen level of the microenvironment in addition to any number of sensors that monitor the temperature control loop values for an infant care station.

In some examples, the alert can be based on the difference between the environmental characteristic and the sealed measurement combined with a difference between the environmental characteristic and an ambient measurement. The ambient measurement can be obtained from an ambient sensor located outside of the microenvironment as described in greater detail above in relation to FIG. 3. In some examples, the unsealed pressure of a microenvironment of an infant care station can be lower than the sealed pressure of the infant care station, and yet can be higher than the ambient pressure level around the infant care station. The method 400 can include, in some examples, determining if one or more access points are unsealed based on whether the pressure of a microenvironment is closer to a sealed measurement or an ambient measurement.

Still at block 408, in some examples, the alert can indicate a particular access point that is experiencing an access point sealing issue. For example, the method 400 can include determining the rate of change of pressure within a microenvironment. A slower rate of change can indicate one or more porthole doors that are unsealed. A faster rate of change of the pressure level can indicate an open or closed canopy in an infant care station. In some examples, the method 400 can include obtaining predetermined rates of change in pressure, or any other suitable environmental characteristic, corresponding to one or two unsealed porthole doors, an unsealed canopy, or any other access points. The method 400 can include generating an alert that indicates the specific access points that are likely unsealed based on the predetermined rates of change in pressure or changes in other environmental characteristics. For example, the method 400 can include determining if one porthole door is unsealed with a sealed canopy, two porthole doors are unsealed with a sealed canopy, two porthole doors are sealed with an unsealed canopy, or any combination thereof.

In some examples, the alert can also include the sensor data detected from one or more microenvironment sensors such that the alert can indicate the pressure sensor values, humidity values, oxygen level values, temperature control loop values, and the like. The sensor data from the alert can be displayed by a display device coupled to an infant care station or a remote device that received the alert from the infant care station.

The process flow diagram of method 400 of FIG. 4 is not intended to indicate that all of the operations of blocks 402-408 of the method 400 are to be included in every example. Additionally, the process flow diagram of method 400 of FIG. 2 describes a possible order of executing operations. However, it is to be understood that the operations of the method 400 can be implemented in various orders or sequences. In addition, in some examples, the method 400 can also include fewer or additional operations. For example, the method 400 can include determining when a canopy of an infant care station is transitioned from an open position to a closed position. In the open position, the canopy does not contact all four side walls so that the

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microenvironment is exposed to the surrounding environment. In the closed position, the canopy contacts all four side walls of the infant care station to seal the microenvironment from the surrounding environment. When the canopy is in a closed position, a radiant heater can be turned off. In some examples, the method 400 can determine that the canopy is in a closed position by monitoring the environmental characteristic to determine an increase in a pressure value of the microenvironment, a temperature control loop value indicating that the heater is still in operation, or the like. The method 400 can generate an alert in response to detecting the heater of the infant care station is still operating with the canopy in a closed position. For example, the method 400 can include generating an alert when a pressure value from a microenvironment falls below a predetermined threshold and turning off the power to a heater of the infant care station.

Additionally, in some examples, the method 400 can include detecting a rate of change of an environmental characteristic and determining that the rate of change is below a predetermined threshold, which can indicate an issue with one or more equipment access points. For example, a gradual loss of pressure below a threshold value can indicate a damaged seal around one or more hoses, cables, and the like, that are placed through access point holes in the walls of an infant care station. In some examples, the method 400 can include generating an alert that indicates a damaged equipment access point in response to detecting a change in an environmental characteristic that is below a predetermined threshold.

FIG. 5 is a block diagram of an example of a computing device that can detect an access point sealing issue of an infant care station. The computing device 500 may be, for example, an infant care station device, such as an incubator, a warmer, or a device that provides features of both an incubator and a warmer, a laptop computer, a desktop computer, a tablet computer, or a mobile phone, among others. The computing device 500 may include a processor 502 that is adapted to execute stored instructions, as well as a memory device 504 that stores instructions that are executable by the processor 502. The processor 502 can be a single core processor, a multi-core processor, a computing cluster, or any number of other configurations. The memory device 504 can include random access memory, read only memory, flash memory, or any other suitable memory systems. The instructions that are executed by the processor 502 may be used to implement a method that can detect an open or unsealed access point of an infant care station, as described in greater detail above in relation to FIG. 4.

The processor 502 may also be linked through the system interconnect 506 (e.g., PCI, PCI-Express, NuBus, etc.) to a display interface 508 adapted to connect the computing device 500 to a display device 510. The display device 510 may include a display screen that is a built-in component of the computing device 500. The display device 510 may also include a computer monitor, television, or projector, among others, that is externally connected to the computing device 500. The display device 510 can include light emitting diodes (LEDs), and micro-LEDs, among others.

The processor 502 may be connected through a system interconnect 506 to an input/output (I/O) device interface 512 adapted to connect the computing device 500 to one or more I/O devices 514. The I/O devices 514 may include, for example, a keyboard and a pointing device, wherein the pointing device may include a touchpad or a touchscreen, among others. The I/O devices 514 may be built-in compo-

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nents of the computing device 500, or may be devices that are externally connected to the computing device 500.

In some embodiments, the processor 502 may also be linked through the system interconnect 506 to a storage device 516 that can include a hard drive, an optical drive, a USB flash drive, an array of drives, or any combinations thereof. In some embodiments, the storage device 516 can include any suitable applications. In some embodiments, the storage device 516 can include an access point manager 518. In some embodiments, the access point manager 518 can obtain a sealed measurement for an infant care station with at least one access point in a sealed position, obtain the environmental characteristic from one or more sensors, and determine a difference between the environmental characteristic and the sealed measurement. The access point manager 518 can also generate an alert indicating an access point sealing issue based on the difference. In some examples, the alert can also indicate that maintenance is to be scheduled for the access point and provide the alert using any suitable user interface or display device. For example, the alert can indicate that a latch or a gasket of a porthole may be malfunctioning, which prevents a porthole door from sealing. The alert can also indicate that a heater is not turning off when an infant care station transitions from a warmer with an open canopy to an incubator with a closed canopy. In some examples, the access point manager 518 can be stored in storage 516 or within memory 504 accessible by the processor 102, among others.

The access point manager 518 can also obtain an ambient measurement from one or more ambient sensors for an environment proximate to an infant care station, obtain an environmental characteristic from one or more microenvironment sensors proximate to at least one access point of the infant care station, determine a difference between the environmental characteristic and the ambient measurement exceeds a predetermined threshold, and generate an alert indicating an access point sealing issue in the infant care station based on the difference exceeding the predetermined threshold. In some examples, the alert can be based on the difference between the environmental characteristic and the ambient measurement, based on the difference between the environment characteristic and a sealed measurement, or a combination thereof. In some examples, the access point manager 518 can generate a first alert in response to detecting a difference between a sealed measurement and an environmental characteristic. The access point manager 518 can also generate a second alert in response to detecting a difference between the ambient measurement and the environmental characteristic that exceeds an ambient threshold. The ambient threshold can indicate a maximum difference between the ambient measurement and the environmental characteristic before an alert is generated.

In some examples, the display device 510 can provide a user interface that indicates data from the alert such as sensor data from the microenvironment sensors, and the like. The display device 510 can also provide a visual representation of an infant care station, wherein the visual representation indicates which of the access points of the infant care station are in an unexpected sealed or unsealed position. For example, the display device 510 can provide a visual representation indicating an open porthole door, a closed canopy with a heater still generating heat for the microenvironment, or the like.

In some examples, the access point manager 518 can obtain proximity sensor data from one or more cameras, proximity sensors, and the like. The access point manager 518 can prevent generating or providing an alert if a user is

detected in the proximity sensor data within a predetermined distance from an infant care station. For example, the access point manager **518** can obtain, detect, or otherwise receive proximity sensor data that indicates the presence of a user near an infant care station. In some examples, if the access point manager **518** detects an unsealed or open access point while a user is proximate to the infant care station, the access point manager **518** can prevent displaying or otherwise providing an alert. In some examples, the access point manager **518** can delay providing an alert for a predetermined period of time or delay providing an alert until a user is no longer in proximate to the infant care station.

In some examples, a network interface controller (also referred to herein as a NIC) **520** may be adapted to connect the computing device **500** through the system interconnect **506** to a network **522**. The network **522** may be a cellular network, a radio network, a wide area network (WAN), a local area network (LAN), or the Internet, among others. The network **522** can enable data, such as alerts, among other data, to be transmitted from the computing device **500** to remote computing devices, remote display devices, remote user interfaces, and the like.

It is to be understood that the block diagram of FIG. **5** is not intended to indicate that the computing device **500** is to include all of the components shown in FIG. **5**. Rather, the computing device **500** can include fewer or additional components not illustrated in FIG. **5** (e.g., additional memory components, embedded controllers, additional modules, additional network interfaces, etc.). Furthermore, any of the functionalities of the access point manager **518** may be partially, or entirely, implemented in hardware and/or in the processor **502**. For example, the functionality may be implemented with an application specific integrated circuit, logic implemented in an embedded controller, or in logic implemented in the processor **502**, among others. In some embodiments, the functionalities of the access point manager **518** can be implemented with logic, wherein the logic, as referred to herein, can include any suitable hardware (e.g., a processor, among others), software (e.g., an application, among others), firmware, or any suitable combination of hardware, software, and firmware.

FIG. **6** depicts a non-transitory machine-executable medium with instructions that can detect an access point sealing issue in an infant care station. The non-transitory, machine-readable medium **600** can cause a processor **602** to implement the functionalities of method **400**. For example, a processor of an infant care station, a host device, a computing device (such as processor(s) **502** of computing device **500** of FIG. **5**), or any other suitable device, can access the non-transitory, machine-readable media **600**.

In some examples, the non-transitory, machine-readable medium **600** can include instructions to execute an access point manager **518**. For example, the non-transitory, machine-readable medium **600** can include instructions that cause the processor **602** to obtain a sealed measurement for an infant care station with at least one access point in a sealed position, obtain the environmental characteristic from one or more sensors, and determine a difference between the environmental characteristic and the sealed measurement. The non-transitory, machine-readable medium **600** can also include instructions that cause the processor **602** generate an alert indicating an access point sealing issue based on the difference. In some examples, the non-transitory, machine-readable medium **600** can include instructions to implement any combination of the techniques of the method **400** described above.

For example, the machine-readable medium **600** can also include instructions to obtain an ambient measurement from one or more ambient sensors for an environment proximate to an infant care station, obtain an environmental characteristic from one or more microenvironment sensors proximate to at least one access point of the infant care station, determine a difference between the environmental characteristic and the ambient measurement exceeds a predetermined threshold, and generate an alert indicating an access point sealing issue in the infant care station based on the difference exceeding the predetermined threshold.

As used herein, an element or step recited in the singular and proceeded with the word “a” or “an” should be understood as not excluding plural of said elements or steps, unless such exclusion is explicitly stated. Furthermore, references to “one embodiment” of the present invention are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Moreover, unless explicitly stated to the contrary, embodiments “comprising,” “including,” or “having” an element or a plurality of elements having a particular property may include additional such elements not having that property. The terms “including” and “in which” are used as the plain-language equivalents of the respective terms “comprising” and “wherein.” Moreover, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements or a particular positional order on their objects.

Embodiments of the present disclosure shown in the drawings and described above are example embodiments only and are not intended to limit the scope of the appended claims, including any equivalents as included within the scope of the claims. Various modifications are possible and will be readily apparent to the skilled person in the art. It is intended that any combination of non-mutually exclusive features described herein are within the scope of the present invention. That is, features of the described embodiments can be combined with any appropriate aspect described above and optional features of any one aspect can be combined with any other appropriate aspect. Similarly, features set forth in dependent claims can be combined with non-mutually exclusive features of other dependent claims, particularly where the dependent claims depend on the same independent claim. Single claim dependencies may have been used as practice in some jurisdictions require them, but this should not be taken to mean that the features in the dependent claims are mutually exclusive.

What is claimed is:

1. An infant care station comprising:

one or more sensors to detect an environmental characteristic of the infant care station;
access points to access a microenvironment of the infant care station; and

a processing system programmed to:

obtain a sealed measurement for the infant care station with the access points in sealed positions;
obtain the environmental characteristic from the one or more sensors;
determine a difference between the environmental characteristic and the sealed measurement exceeds a predetermined threshold; and
generate an alert indicating an access point sealing issue based on the difference and indicate which of the access points is experiencing the sealing issue.

2. The infant care station of claim 1, wherein the access points comprise a canopy, a porthole door, an equipment access point, or a combination thereof.

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3. The infant care station of claim 1, wherein the processing system is programmed to transmit the alert to an external computing device or a remote display device.

4. The infant care station of claim 1, wherein the processing system is programmed to display the alert in a user interface coupled to the system.

5. The infant care station of claim 1, wherein the environmental characteristic comprises a pressure value, a humidity value, an oxygen value, a temperature control loop value, or a combination thereof.

6. The infant care station of claim 5, wherein the one or more sensors comprise a pressure sensor, an oxygen sensor, a humidity sensor, a control loop sensor, or a combination thereof.

7. The infant care station of claim 5, wherein the one or more sensors are coupled to an enclosure of the system, wherein the enclosure supports the microenvironment.

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8. The infant care station of claim 1, wherein the alert comprises an audio feedback, haptic feedback, or visual feedback representing a message that at least one of the access points is unsealed.

9. The infant care station of claim 1, further comprising an ambient sensor, wherein the ambient sensor is to detect an ambient measurement, and wherein the processing system is programmed to generate a second alert in response to detecting a difference between the ambient measurement and the environmental characteristic that exceeds an ambient threshold.

10. The infant care station of claim 1, wherein the one or more sensors comprise a pressure sensor within the microenvironment and the environmental characteristic comprises a pressure value obtained from the microenvironment by the pressure sensor.

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